Assignment: Nonlinear System

Name: 유채은 **ID: 22200476**

Academic Integrity Declaration

Check the statement that applies and sign your name.

Check	Statement
V	I have **NOT** used LLM (ChatGPT, etc), any online materials, or other students' reports/codes to complete this assignment.
	I have **NOT completed this assignment entirely on my own**. (Programming-related assistance, *including debugging*)

If you received any assistance **that violates the class policy**, you must explain it in detail. A penalty will be applied depending on the **extent and nature of the assistance**.

- For being honest with the assistance, you may get penalty within this assistance score.
- However, Failure to disclose the assistance, you will get Grade F.

Examples:

- *Share Link or Screenshots of LLM prompt,*
- *Include Program related assistance, including debugging*)

  ** You don't need to include LLM prompt for inquiring mathematical concepts*
- *Other Explanation of assistance : who/how/what extent etc*

By signing below, I confirm that I have honestly and truthfully answered the academic integrity declaration.

Date: 2025.12.12

Sign: *유채은*

What you need to submit

- **Submit**: Report & Source files as one ZIP file
  - Assignment_NonlinearSys_22200476.zip
- **Report:** Pseudocode, output result and source code as instructed
- **Source code files:**
  - Assignment_NonlinearSystem _22200476.cpp
  - myNP_22200476.h, myNP_22200476.cpp, myMatrix_22200476.h, myMatrix_22200476.cpp

Part 0: Follow Tutorial in MATLAB

[Read tutorial explanation:]

Part 1: Newton-Raphson for Nonlinear System

[Download the template source file]

Problem 1: (10pt)

Solve for x,y

$$f_1(x,y) = y - \frac{1}{2}\left(e^{x/2} + e^{-x/2}\right)=0$$

$$f_2(x,y) = 9x^2 + 25y^2 - 225=0$$

ANS:

x=__3.031157______, y=_2.385865__________

Procedure

a) Write down a pseudocode for solving a system of nonlinear equations using Newton-Raphson

---

```
F =-1*F
H = J\(-F)
Z = Z + H
```

---

b) Create C/C++ function

```
Matrix myFuncEx1(Matrix _vecZ);     // returns F(z) (nx1)
Matrix myJacobEx1(Matrix _vecY);     //   returns Matrix J   (nxn)

 Matrix nonlinearSys(Matrix myFuncs(Matrix _vecZ),
                     Matrix Jacob(Matrix _vecZ), Matrix _vecZinit,
                      double tol);
```

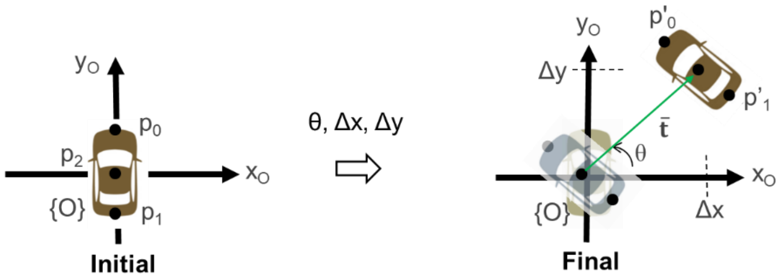
```
// Initialization
int n = _Z0.rows;
double loss = 10;
Matrix J = zeros(n, n);
Matrix F = zeros(n, 1);
Matrix H = zeros(n, 1);
Matrix Z = zeros(n, 1);

Matrix L = eye(n, n);
Matrix U = zeros(n, n);
Matrix P = eye(n, n);

// Initial condition
Z = _Z0;

for (int k = 0; k < 20; k++) {
        F = Funcs(Z);
        J = Jacob(Z);

        // F =-1*F;
        F = smultMat(F, (-1));
```

```
        // Solve for H in    JH = (-F) -> obtain 'H'
        LUdecomp(&J, &L, &U, &P);
        solveLU(&L, &U, &P, &F, &H);


        // Z = Z + H;
        Z = addMat(Z, H);

        // After Z update, print loss
        F = Funcs(Z);

        // loss = norm2(F);
        Matrix FT = transpose(F);
        loss = sqrt(vTv(FT, F));


        printf("iter =%d \t x=%0.3f \t y=%0.3f \t loss=%0.3f\n", k, Z.at[0][0], Z.at[1][0],
loss);

        if (loss < tol) {
                printf("Early Termination\n\n\r");
                return Z;
        }
}
return Z;
```

Problem 2: (10pt)

Solve for the 3-DOF transformation, angle (θ) and translation T=[Δx, Δy], to move the vehicle to the new pose.

Read the lecture notes and MATLAB tutorial for help.



p_0=[0,100]
p_1=[0,-100]
p_2=[0, 0]

p'_0=[50, 186.6025]
p'_1=[150, 13.3975]
p'_2=[100, 100]

ANS:

(θ)=__30____[deg] , T=[Δx, Δy]= ___[100, 100]_____ [mm]

```
-----------------------------------------------------------------------------------
                              System of NonLinear

-----------------------------------------------------------------------------------
problem 1.

iter =0          x=3.139         y=2.400         loss=7.674
iter =1          x=3.034         y=2.385         loss=0.104
iter =2          x=3.031         y=2.386         loss=0.000
Early Termination

Z =
       3.031157
       2.385865

--------------------------------------------------------------------------------
problem 2.

iter =0          th=0.500        x=100.000       y=86.602        loss=12.583
iter =1          th=0.523        x=100.000       y=99.968        loss=0.031
iter =2          th=0.524        x=100.000       y=100.000       loss=0.000
Early Termination

Z2 =
       0.523599
     100.000000
      99.999958

theta = 30.00[deg]

Press any key to continue . . .
```